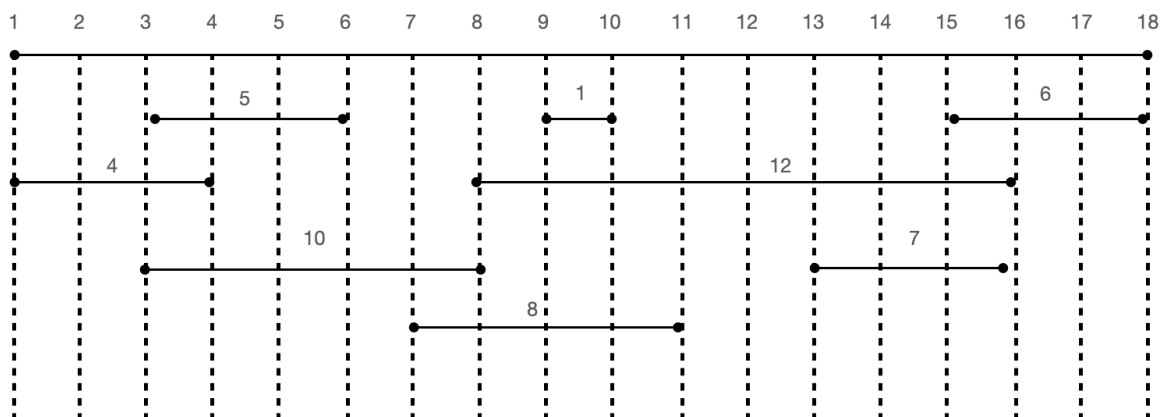


1. Assembly and de Bruijn graph. Given the following reads, draw the de Bruijn graph (3-mer as node), and write down what is the whole sequence.

- 1) \$ATGC
- 2) TGCAT
- 3) ATGCT
- 4) GCTAG
- 5) CTAGC
- 6) TAGCA
- 7) AGCA\$
- 8) ATGCA
- 9) GCATG
- 10) TGCTA
- 11) CATGC

2. Here is the position and weight of all exon segments in an abstract sequence of length 18. Please find a maximum set of non-overlapping putative exons based on brute force or dynamic programming.



3. Please generate a dynamic programming matrix for sequence alignment and find its optimum alignment score.

Sequence_A: GATTCATA

Sequence_B: GGATCGA

Scoring system: +5 for a match; -2 for a mismatch; -6 for each indel

4. Homology Search and Phylogenetic Analysis of MLH1

Objective:

Use BLAST, multiple sequence alignment, and phylogenetic tree construction to explore the evolutionary relationship of the **human MLH1** gene across species.

Tasks:

1. Use **BLASTp** to search homologous protein sequences of MLH1 from the NCBI .
2. Select representative homologs from diverse species (about 10–15 sequences).
3. Download the protein sequences in FASTA format and combine them into one file.
4. Perform **multiple sequence alignment** (MSA) using your preferred alignment tool
5. Construct a **phylogenetic tree** using your aligned sequences.
6. Visualize and annotate the tree.

Submit:

- The alignment file (**.fasta**),
- The phylogenetic tree (image), and
- A short summary (within one page) describing your findings and observations.